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1 Gauss-Legendre Quadrature on the Unit Sphere

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1.2 1. Objective

The goal is to construct a numerical quadrature rule for integrating functions over the unit sphere using a product quadrature rule consisting of:

- **Gauss-Legendre quadrature** in the variable $\mu = \cos \theta$
 - **Periodic trapezoidal quadrature** in the azimuthal variable ϕ
-

1.3 2. Spherical Coordinates

A point on the unit sphere is parameterized by

$$\mathbf{r}(\theta, \phi) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta),$$

with

$$0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi.$$

The surface element is

$$d\Omega = \sin \theta \, d\theta \, d\phi.$$

Therefore,

$$\int_{S^2} f \, d\Omega = \int_0^\pi \int_0^{2\pi} f(\theta, \phi) \sin \theta \, d\phi \, d\theta.$$

1.4 3. Transformation to $\mu = \cos \theta$

Introduce

$$\mu = \cos \theta.$$

Then

$$d\mu = -\sin \theta \, d\theta,$$

and the limits become

$$\theta = 0 \Rightarrow \mu = 1, \quad \theta = \pi \Rightarrow \mu = -1.$$

Hence,

$$\int_0^\pi f(\theta) \sin \theta \, d\theta = \int_{-1}^1 f(\mu) \, d\mu.$$

Thus,

$$\boxed{\int_{S^2} f d\Omega = \int_{-1}^1 \int_0^{2\pi} f(\mu, \phi) d\phi d\mu.}$$

1.5 4. Gauss-Legendre Quadrature in μ

An N_μ -point Gauss-Legendre rule is

$$\int_{-1}^1 g(\mu) d\mu \approx \sum_{i=1}^{N_\mu} w_i^\mu g(\mu_i).$$

The points are the roots of

$$P_{N_\mu}(\mu_i) = 0,$$

with weights

$$w_i^\mu = \frac{2}{(1 - \mu_i^2) \left[P'_{N_\mu}(\mu_i) \right]^2}.$$

The rule integrates polynomials exactly through degree

$$\boxed{2N_\mu - 1}.$$

1.6 5. Periodic Trapezoidal Rule in ϕ

Choose N_ϕ equally spaced points

$$\phi_j = \frac{2\pi j}{N_\phi}, \quad j = 0, \dots, N_\phi - 1.$$

The weights are

$$w_j^\phi = \frac{2\pi}{N_\phi}.$$

Therefore,

$$\int_0^{2\pi} h(\phi) d\phi \approx \sum_{j=0}^{N_\phi-1} w_j^\phi h(\phi_j).$$

1.7 6. Product Quadrature Rule

Combining both rules gives

$$Q[f] = \sum_{i=1}^{N_\mu} \sum_{j=0}^{N_\phi-1} w_i^\mu w_j^\phi f(\mu_i, \phi_j).$$

Equivalently,

$$Q[f] = \frac{2\pi}{N_\phi} \sum_{i=1}^{N_\mu} w_i^\mu \sum_{j=0}^{N_\phi-1} f(\mu_i, \phi_j).$$

1.8 7. Quadrature Points on the Sphere

Each pair (μ_i, ϕ_j) corresponds to

$$x_{ij} = \sqrt{1 - \mu_i^2} \cos \phi_j,$$

$$y_{ij} = \sqrt{1 - \mu_i^2} \sin \phi_j,$$

$$z_{ij} = \mu_i.$$

Hence,

$$\mathbf{r}_{ij} = \left(\sqrt{1 - \mu_i^2} \cos \phi_j, \sqrt{1 - \mu_i^2} \sin \phi_j, \mu_i \right).$$

The product weight is

$$W_{ij} = w_i^\mu w_j^\phi.$$

1.9 8. Spherical Harmonics

The complex spherical harmonics are written as

$$Y_\ell^m(\mu, \phi) = \Omega_{\ell m} P_\ell^m(\mu) e^{im\phi},$$

where

$$\Omega_{\ell m} = \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - m)!}{(\ell + m)!}}.$$

The indices satisfy

$$\ell = 0, 1, 2, \dots, \quad -\ell \leq m \leq \ell.$$

For the standard normalization,

$$\int_{S^2} Y_\ell^{m*} Y_{\ell'}^{m'} d\Omega = \delta_{\ell\ell'} \delta_{mm'}.$$

1.10 9. Resolution Requirement

A sufficient resolution for spherical-harmonic orthogonality up to degree L is

$$\boxed{N_\mu \geq L + 1}, \quad \boxed{N_\phi \geq 2L + 1}.$$

The Gauss-Legendre rule with N_μ points integrates polynomials through degree

$$2N_\mu - 1.$$

Therefore, choosing

$$N_\mu \geq L + 1$$

is sufficient to integrate the required polar polynomial products.

For the azimuthal direction, the spherical harmonics contain Fourier modes

$$e^{im\phi}.$$

Choosing

$$N_\phi \geq 2L + 1$$

provides sufficient resolution for products of spherical harmonics up to degree L .

1.11 10. Constant Function Test

The exact area of the unit sphere is

$$\int_{S^2} 1 d\Omega = 4\pi.$$

For the quadrature,

$$Q[1] = \left(\sum_{i=1}^{N_\mu} w_i^\mu \right) \left(\sum_{j=0}^{N_\phi-1} w_j^\phi \right).$$

Gauss-Legendre gives

$$\sum_i w_i^\mu = 2,$$

and the trapezoidal rule gives

$$\sum_j w_j^\phi = 2\pi.$$

Therefore,

$$Q[1] = 2(2\pi) = 4\pi.$$

This confirms that the quadrature correctly integrates a constant function over the sphere.

1.12 11. Algorithm

1.12.1 Input

Parameter	Description
N_μ	Number of Gauss-Legendre nodes in μ
N_ϕ	Number of equally spaced nodes in ϕ
tolerance	Convergence tolerance for Newton's method

1.12.2 Output

Quantity	Description
$\{\mu_i, w_i^\mu\}_{i=1}^{N_\mu}$	Gauss-Legendre nodes and weights
$\{\phi_j, w_j^\phi\}_{j=0}^{N_\phi-1}$	Trapezoidal nodes and weights
$\{\mathbf{r}_{ij}\}$	Quadrature points on S^2
$\{W_{ij}\}$	Product quadrature weights

1.12.3 Step 1: Compute Gauss-Legendre Nodes and Weights

For each $i = 1, \dots, N_\mu$, compute the initial Newton guess

$$\mu_i^{(0)} = \cos \left(\pi \frac{i - 1/4}{N_\mu + 1/2} \right).$$

Evaluate the Legendre polynomial using the recurrence

$$(k+1)P_{k+1}(\mu) = (2k+1)\mu P_k(\mu) - kP_{k-1}(\mu),$$

with

$$P_0(\mu) = 1, \quad P_1(\mu) = \mu.$$

The derivative is

$$P'_{N_\mu}(\mu) = \frac{N_\mu}{\mu^2 - 1} [\mu P_{N_\mu}(\mu) - P_{N_\mu-1}(\mu)].$$

Newton's method is

$$\mu^{(k+1)} = \mu^{(k)} - \frac{P_{N_\mu}(\mu^{(k)})}{P'_{N_\mu}(\mu^{(k)})}.$$

Continue until

$$|\mu^{(k+1)} - \mu^{(k)}| < \text{tolerance}.$$

After convergence, calculate the weight

$$w_i^\mu = \frac{2}{(1 - \mu_i^2) [P'_{N_\mu}(\mu_i)]^2}.$$

1.12.4 Step 2: Compute Trapezoidal Nodes and Weights

For

$$j = 0, \dots, N_\phi - 1,$$

compute

$$\phi_j = \frac{2\pi j}{N_\phi},$$

and

$$w_j^\phi = \frac{2\pi}{N_\phi}.$$

1.12.5 Step 3: Construct Quadrature Points on the Sphere

For each i , calculate

$$\rho_i = \sqrt{1 - \mu_i^2}.$$

Then, for each j ,

$$x_{ij} = \rho_i \cos \phi_j,$$

$$y_{ij} = \rho_i \sin \phi_j,$$

$$z_{ij} = \mu_i.$$

Store

$$\mathbf{r}_{ij} = (x_{ij}, y_{ij}, z_{ij}).$$

1.12.6 Step 4: Compute Product Weights

For each i, j ,

$$W_{ij} = w_i^\mu w_j^\phi.$$

Since the azimuthal weights are equal,

$$W_{ij} = w_i^\mu \frac{2\pi}{N_\phi}.$$

1.12.7 Step 5: Apply Quadrature Rule

For a function $f(\mu, \phi)$, calculate

$$Q[f] = \sum_{i=1}^{N_\mu} \sum_{j=0}^{N_\phi-1} W_{ij} f(\mu_i, \phi_j).$$

Equivalently,

$$Q[f] = \frac{2\pi}{N_\phi} \sum_{i=1}^{N_\mu} w_i^\mu \sum_{j=0}^{N_\phi-1} f(\mu_i, \phi_j).$$

1.13 12. Results

All four checks were run with $N_\mu = 6$ and $N_\phi = 13$. Each figure below corresponds to one demonstration case, presented in the same order as the assignment requirements.

Product quadrature nodes on the unit sphere
 $n_{\mu}=6$, $n_{\phi}=13$, azimuthal='trapezoidal' (78 nodes)

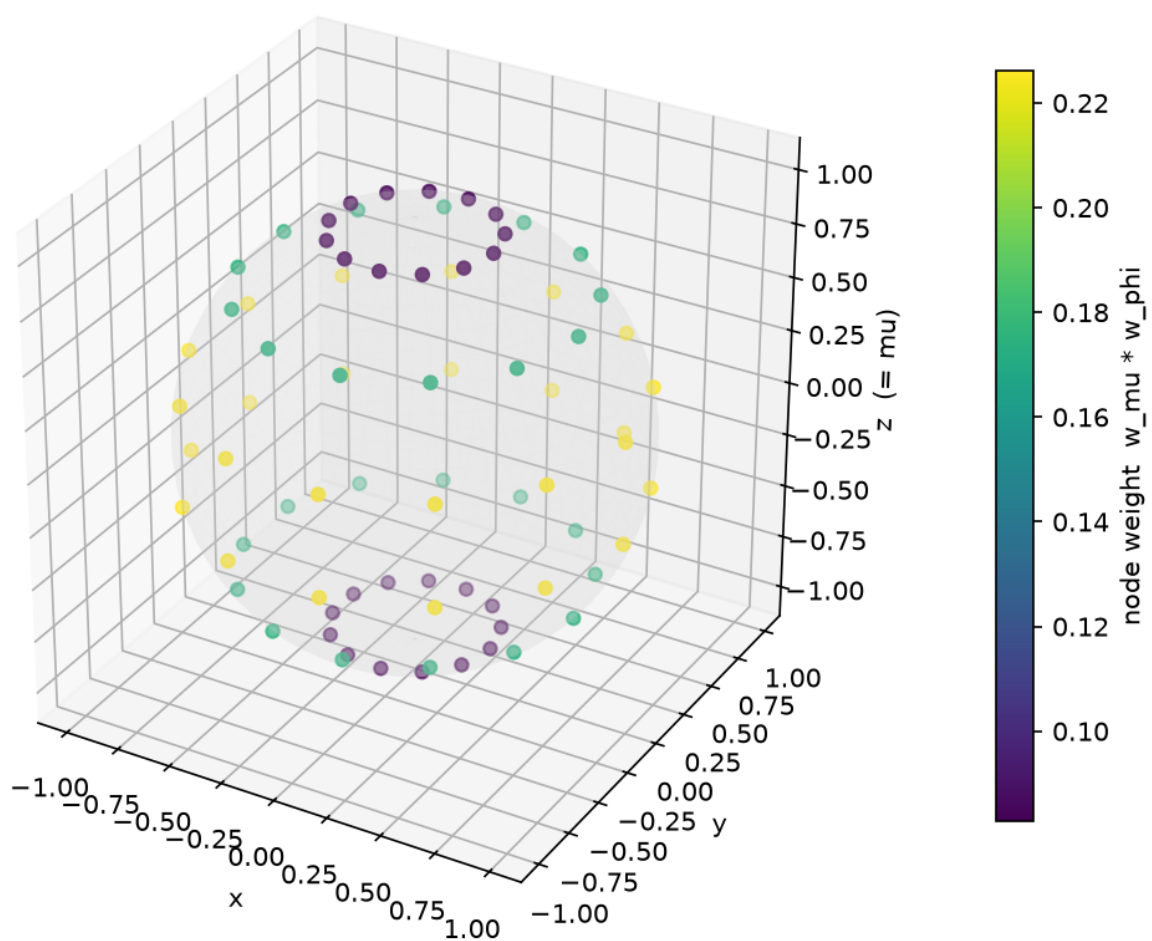


Figure 1: Product quadrature nodes on the unit sphere

1.13.1 12.1 Product Quadrature Nodes on the Unit Sphere

The 3-D visualization shows all $6 \times 13 = 78$ product quadrature nodes placed exactly on the unit sphere, colored by their product weight $W_{ij} = w_i^\mu w_j^\phi$. The nodes cluster more densely near the poles (as expected from Gauss-Legendre) and are uniformly spaced in the azimuthal direction. The sum of all weights equals $4\pi \approx 12.566371$ with error $\sim 10^{-15}$, confirming the constant-function test.

Figure 1: Product quadrature nodes on the unit sphere ($N_\mu = 6$, $N_\phi = 13$, trapezoidal).

1.13.2 12.2 Polynomial Exactness in μ (Gauss-Legendre)

The plot shows the absolute error in integrating μ^k for $k = 0, \dots, 14$ with $N_\mu = 6$. Within the guaranteed range $k \leq 2N_\mu - 1 = 11$ (blue curve), errors sit at machine precision ($\sim 10^{-16}$), demonstrating exactness. Beyond $k = 11$ (red curve), errors jump to $\sim 10^{-3}$ - 10^{-2} , confirming that the rule fails past the theoretical degree limit (dashed vertical line).

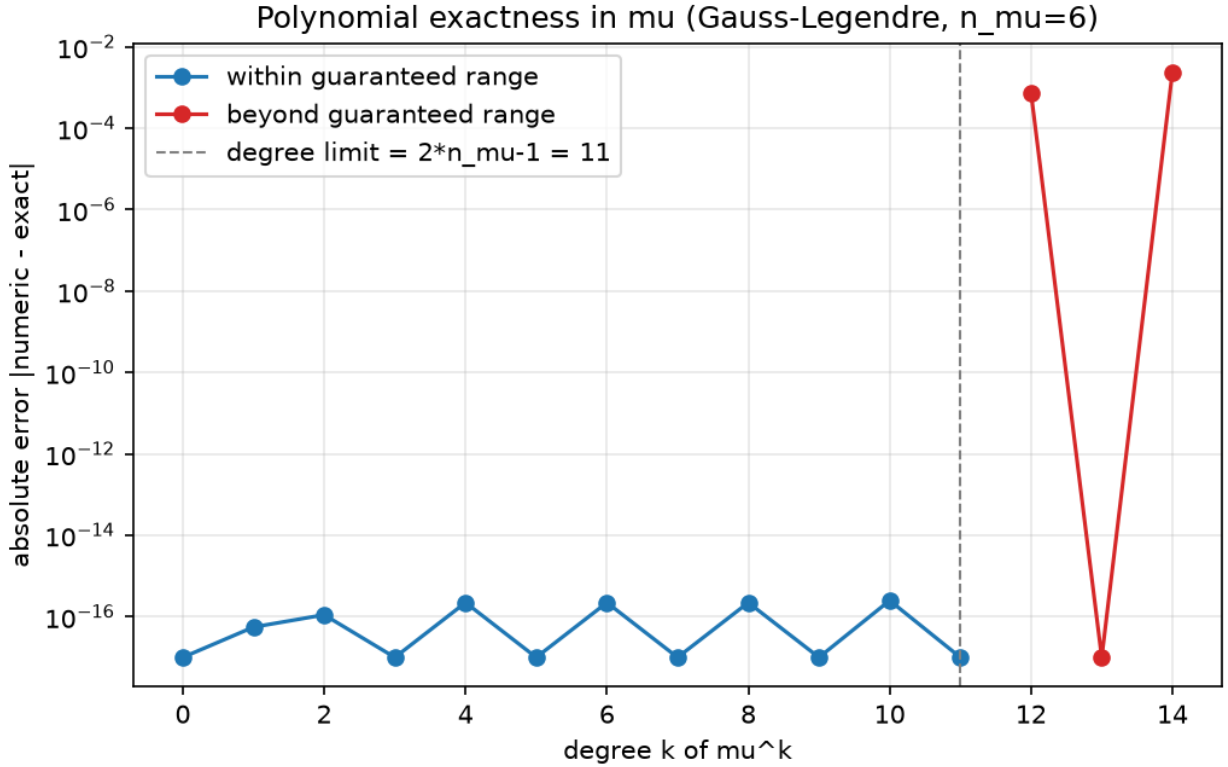


Figure 2: Gauss-Legendre polynomial exactness in μ

Figure 2: Gauss-Legendre polynomial exactness in μ ($N_\mu = 6$).

1.13.3 12.3 Resolution Limit for Spherical Harmonics

This plot sweeps the diagonal self-overlap $\int |Y_n^n|^2 d\Omega$ for $n = 0, \dots, 11$ using $N_\mu = 6$, $N_\phi = 13$. Machine-precision exactness holds for $n \leq 5 = N_\mu - 1$; the error jumps at $n = 6 = N_\mu$ (dashed line) to $\sim 10^{-3}$ and grows to $\sim 5 \times 10^{-2}$ at $n = 11$ (red point). This confirms the resolution condition $N_\mu \geq L + 1$ and demonstrates the predicted failure beyond it.

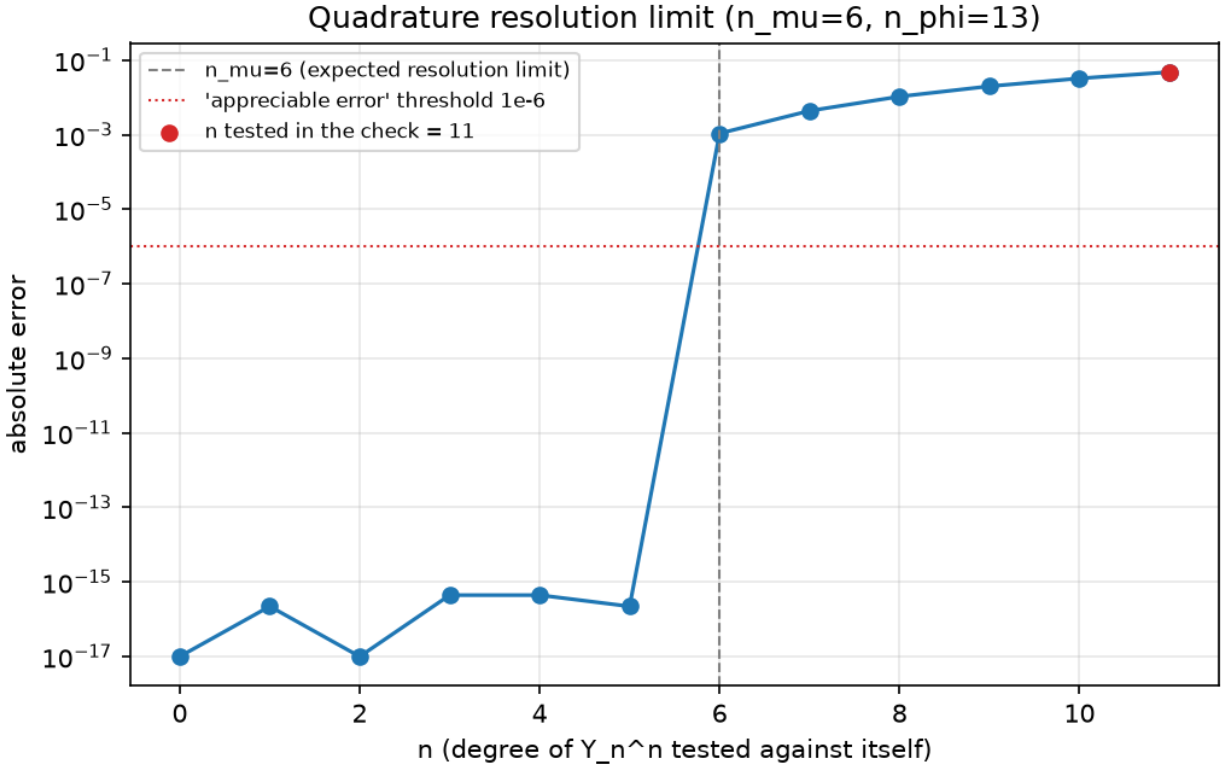


Figure 3: Resolution limit for spherical harmonics

Figure 3: Resolution limit for spherical harmonics ($N_\mu = 6$, $N_\phi = 13$).

1.13.4 12.4 Gauss-Chebyshev ↔ Trapezoidal Equivalence (Bonus)

The left polar plot shows the two azimuthal node sets for $N_\phi = 13$: trapezoidal nodes (blue dots) and Gauss-Chebyshev nodes (orange crosses), which sit exactly halfway between the trapezoidal nodes (constant shift $\pi/N_\phi \approx 0.241661$). The right plot shows both rules integrate $e^{ik\phi}$ to machine precision for $k \leq 12 = N_\phi - 1$; at $k = 13 = N_\phi$ both fail with equal magnitude $|2\pi| \approx 6.283$ but opposite signs ($+6.283$ vs. -6.283), confirming the aliasing sign factor $(-1)^j$. This numerically verifies the mathematical equivalence between the two azimuthal rules.

Figure 4: Gauss-Chebyshev ↔ trapezoidal equivalence ($N_\phi = 13$).

1.14 12. How to Reproduce

```
# Basic run
python3 run.py
```

```
# Individual case (edit input.txt first)
python3 run.py
```

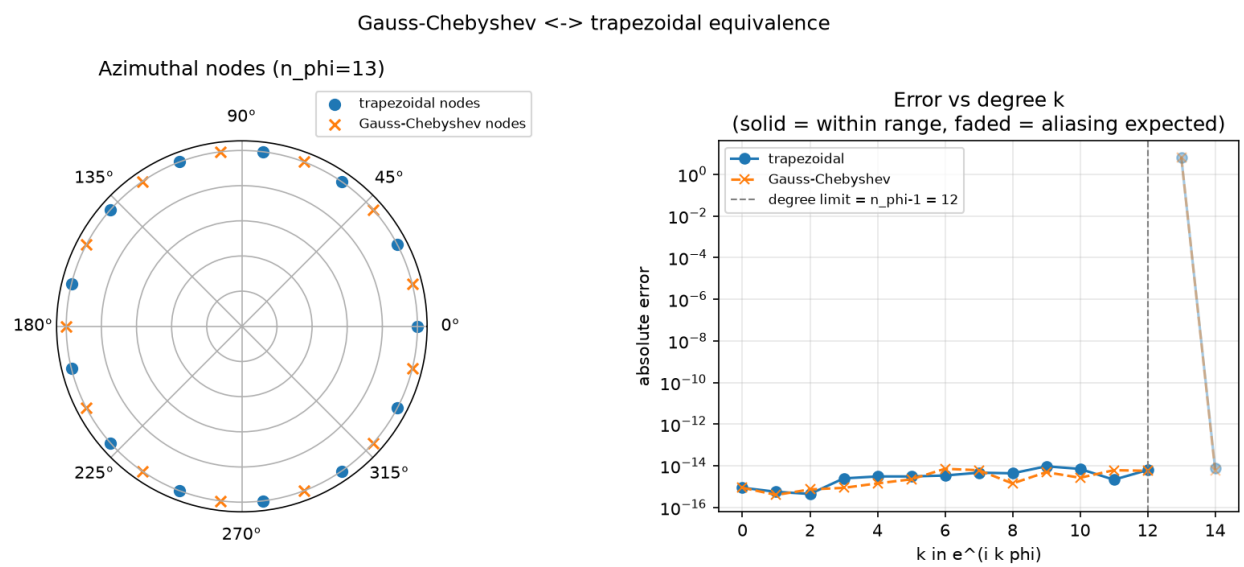


Figure 4: Gauss-Chebyshev vs trapezoidal equivalence